SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I B.Pardhu Arjun (192525245) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Pardhu Arjun

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Solution:

SCENARIO 1:

Smart Campus Network Subnetting
Scenario Understanding
A Smart Campus has been assigned the IP address block 192.168.100.0/24. Different departments require different subnet sizes based on the expected number of devices. The objective is to allocate IP addresses efficiently using subnetting while minimizing address wastage.

Given Data

Department	Subnet
School of Computing	/25
School of Electronics	/26
School of Mechanical Engineering	/27

Step 1: Calculate the Subnets
School of Computing

Parameter	Value
Network Address	192.168.100.0
Subnet Mask	255.255.255.128
Broadcast Address	192.168.100.127
Host Range	192.168.100.1 – 192.168.100.126
Usable Hosts	126

School of Electronics

Parameter	Value
Network Address	192.168.100.128
Subnet Mask	255.255.255.192
Broadcast Address	192.168.100.191
Host Range	192.168.100.129 – 192.168.100.190
Usable Hosts	62

School of Mechanical Engineering

Parameter	Value
Network Address	192.168.100.192
Subnet Mask	255.255.255.224
Broadcast Address	192.168.100.223
Host Range	192.168.100.193 – 192.168.100.222
Usable Hosts	30

Remaining Unused Address Range

Starting Address	Ending Address
192.168.100.224	192.168.100.255

Observation Table

Department	Network Address	Host Range	Broadcast Address	Usable Hosts
Computing	192.168.100.0/25	192.168.100.1–126	192.168.100.127	126
Electronics	192.168.100.128/26	192.168.100.129–190	192.168.100.191	62
Mechanical	192.168.100.192/27	192.168.100.193–222	192.168.100.223	30

Decision
The subnet allocation efficiently divides the available 192.168.100.0/24 network among all departments. Each department receives sufficient host addresses, and the remaining IP addresses (192.168.100.224–192.168.100.255) are reserved for future expansion.

SCENARIO 2:

Multinational Software Company
Scenario Understanding
A software company has received the private network 10.10.0.0/16. Each department has different networking requirements. Appropriate subnet masks are assigned to maximize IP utilization while supporting future growth.

Given Data

Department	Subnet
Software Development	/24
Quality Assurance	/25
Human Resources	/26
Executive Management	/27

Subnet Allocation

Department	Network Address	Host Range	Broadcast Address	Usable Hosts
Software Development	10.10.0.0/24	10.10.0.1–10.10.0.254	10.10.0.255	254
Quality Assurance	10.10.1.0/25	10.10.1.1–10.10.1.126	10.10.1.127	126
Human Resources	10.10.1.128/26	10.10.1.129–10.10.1.190	10.10.1.191	62
Executive Management	10.10.1.192/27	10.10.1.193–10.10.1.222	10.10.1.223	30

Remaining Unused Address Range

Starting Address	Ending Address
10.10.1.224	10.10.255.255

Decision

Variable Length Subnet Masking (VLSM) efficiently allocates addresses according to department size. The remaining address space can be used for future departments or network expansion.

SCENARIO 3:

IPv6 Migration
Scenario Understanding
A smart hospital is migrating from IPv4 to IPv6 to support thousands of medical devices and IoT sensors. IPv6 offers a significantly larger address space than IPv4.

Given IPv6 Address
2001:0db8:abcd:0000:0000:0000:0000:0000/64

IPv6 Analysis

Task	Answer
Compressed Address	2001:db8:abcd::/64
Expanded Address	2001:0db8:abcd:0000:0000:0000:0000:0000
Network Prefix	2001:db8:abcd::/64
Host Bits	64 bits
Total Host Addresses	2^{64} = 18,446,744,073,709,551,616

Decision
IPv6 provides an enormous address space, making it ideal for hospitals with thousands of connected medical devices, IoT sensors, and future smart healthcare applications.

SCENARIO 4:

Routing Table Analysis
Scenario Understanding
A router receives a packet destined for 192.168.2.45. The router checks its routing table to identify the matching subnet and forwards the packet using the appropriate interface.

Routing Table

Network
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24

Routing Decision

Question	Answer
Destination IP	192.168.2.45
Destination Subnet	192.168.2.0/24
Matching Route	192.168.2.0/24
Next-Hop Interface	Interface connected to 192.168.2.0/24
Default Route	0.0.0.0/0

Decision

The router performs a longest-prefix match and forwards the packet through the interface associated with the 192.168.2.0/24 subnet. If no matching route exists, the packet is forwarded using the default route.

SCENARIO 5:

Office LAN Configuration
Scenario Understanding

A company has two LANs connected through a Layer-3 router. Each LAN requires proper network addressing, host allocation, and gateway configuration to ensure reliable communication.

Given Data

LAN	Subnet	Gateway
Administration	192.168.10.0/26	192.168.10.1
Finance	192.168.10.64/26	192.168.10.65

Network Details

LAN	Network Address	Host Range	Broadcast Address	Gateway
Administration	192.168.10.0	192.168.10.1–192.168.10.62	192.168.10.63	192.168.10.1
Finance	192.168.10.64	192.168.10.65–192.168.10.126	192.168.10.127	192.168.10.65

Suggested IP Addresses

Administration LAN

Computer	IP Address
PC1	192.168.10.2
PC2	192.168.10.3
PC3	192.168.10.4

Finance LAN

Computer	IP Address
PC1	192.168.10.66
PC2	192.168.10.67
PC3	192.168.10.68

Decision

The two LANs are correctly subnetted using /26 networks. Each LAN has its own gateway, preventing address conflicts and enabling efficient communication through the Layer-3 router. The design also allows for easy network management and future expansion.